

Metro - Homework 12

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Question 1

(1.a) let X be T_1 space

Claim 1.a.1 every $Y \subset X$ is also T_1

Proof. let $Y \subset X$, take $x \neq y \in Y$, since $x, y \in X$, then exists $U \subset X$ open such that:

$$x \in U, y \notin U$$

by definition, hence, we have:

$$x \in U \cap Y, y \notin U \cap Y$$

therefore, we have $V = U \cap Y$ the desired open nghbr, therefore Y is T_1 as required. $\square$

(1.b)

Claim 1.b.1 if X is normal, and $A \subset X$ a closed set, then A is normal

Proof. let $B_1, B_2 \subset A$ be both disjoint closed sets, since A is closed, then we also have

B_1, B_2 closed in X , therefore, since X is normal, by definition, exists $V_1, V_2 \subset X$

disjoint open such that:

$$B_1 \subset V_1, B_2 \subset V_2$$

Consequently,

$$B_1 = B_1 \cap A \subset V_1 \cap A = U_1 \in \tau_A$$

$$B_2 = B_2 \cap A = V_2 \cap A = U_2 \in \tau_A$$

Hence,

$$B_1 \subset U_1, B_2 \subset U_2$$

$$U_1 \cap U_2 = \emptyset$$

Therefore, A is normal by definition. □

Question 2

let X be a topological space and $A \subset X$ closed set.

define the following relation:

$$x \sim y \iff x = y \vee \{x, y\} \subset A$$

Claim 2.1 if X is T_3 then $X/\sim$ is T_2

Proof. let $[x] \neq [y] \in X/\sim$, then by the definition of the relation we have:

$$(x \neq y) \wedge (x, y \notin A)$$

therefore, since $x \neq y$, and X is T_2 , exists V_x, V_y disjoint open nighbors of x, y respectively, the problem of taking these sets immediately to $X/\sim$ is that they might intersect A and then they are not disjoint in $\tau_{X/\sim}$, hence, assume without loss of generality, $x \notin A$, by the regularity of X , for $x \notin A$, exists U_x , open such that:

$$U_x \cap A = \emptyset$$

Therefore, we take:

$$W_x = V_x \cap U_x, \quad W_y = V_y$$

Both are open, and non empty since they contain x, y respectively, and disjoint since V_x, V_y are disjoint. hence:

$$Z_x = \pi(W_x) = \{[z] : z \in W_x\}$$

$$Z_y = \pi(W_y) = \{[z] : z \in W_y\}$$

Are both open in $X/\sim$ by definition, since $W_x \cap A = \emptyset$, then $[A] \notin Z_x$. and we have

Z_x, Z_y disjoint, since otherwise:

$$[z] \in Z_x \cap Z_y$$

then, since $[z] \neq [A]$ then

$$z \in W_x \cap W_y$$

In contradiction to their disjoint.

hence, we have:

$$[x] \in W_x$$

$$[y] \in W_y$$

and they are open and disjoint, therefore, $X/\sim$ is T_2 as required. $\square$

Question 3

Claim 3.1 let X be regular space and second countable then X is normal.

Proof. let $\{B_n\}_{n=1}^{\infty}$ be a countable basis for X , take $A, B \subset X$ both closed disjoint.

for each $a \in A$, we have $a \notin B$, therefore, by the regularity of X , there exists disjoint

Q_a, W_a open such that:

$$a \in Q_a, B \subset W_a$$

hence, we have:

$$A \subset \bigcup_{a \in A} Q_a$$

since X is second countable, there exists $f: A \rightarrow \mathbb{N}$ such that we have:

$$a \in B_{f(a)} \subset Q_a$$

therefore, we replace the indexing of A with $\{x_n\}_{n=1}^{\infty} \subset f(A)$, and define $U_n = B_{x_n}$

hence we have:

$$A \subset \bigcup_{n=1}^{\infty} U_n$$

and for each n , pick $a_n \in f^{-1}(x_n)$ and define:

$$B \subset W_{a_n} = V_n$$

where a_n is a representative of such $f^{-1}(n)$'s. therefore, we obtain that:

$$U_n \subset Q_{a_n} \implies U_n \cap V_n = \emptyset$$

now define the following:

$$U'_n = U_n \setminus \bigcup_{k=1}^n \overline{V_k}$$
$$V'_n = V_n \setminus \bigcup_{k=1}^n \overline{U_k}$$

which are open sets, since they are open sets minus closed ones.

and we have

$$\overline{U_n} \cap B = \emptyset$$

because otherwise, take $x \in \overline{U_n} \cap B$, then we have:

$$x \in B \implies x \in V_n$$

V_n is an open set, therefore, exists P open nghbr of x such that:

$$x \in P \subset V_n$$

and,

$$x \in \overline{U_n} \implies P \cap U_n \neq \emptyset$$

Therefore,

$$U_n \cap V_n \neq \emptyset$$

in contradiction.

and we have

$$V'_n \cap A = \emptyset$$

hence, U'_n does not remove elements from A , meaning we have

$$A \subset \bigcup U_n \implies A \subset \bigcup U'_n$$

$$B \subset \bigcup V_n \implies B \subset \bigcup V'_n$$

define:

$$U = \bigcup_{n=1}^{\infty} U'_n, \quad V = \bigcup_{n=1}^{\infty} V'_n$$

both of which are open, because they are union of open sets. therefore,

$$U \cap V = \emptyset$$

because, otherwise, $\exists x$ such that:

$$x \in U \implies \exists i \ x \in U'_i$$

$$x \in V \implies \exists j \ x \in V'_j$$

if without loss of generality $i < j$, then we have:

$$x \in V'_j \implies x \in V_j \setminus \bigcup_{k=1}^j \overline{U_k}$$

In particular, we must have:

$$x \notin \overline{U_i}$$

which is a contradiction to $x \in U'_i$.

Consequently, we have

$$A \subset U, B \subset V$$

both open and disjoint as required.

□